Project Smart Trigger Design Documentation for ECE/CS 3992

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***Abstract*—This document describes the design of the Smart Trigger, a modular RFID-based access-control system for the AR-15 platform. A passive 125kHz RFID tag worn by the authorized user is read by an antenna embedded in a custom 3D-printed grip. A Raspberry Pi Pico microcontroller validates the tag against an authorized list and drives a micro stepper motor that moves a blocking member out of the fire control group. The document covers the system architecture, the hardware components and their integration onto a custom PCB, the CAD modeling of the grip and actuator mounts, and the firmware processes that tie the system together. It closes with a look at how the design could grow through an off-device logging server and web application, and what that complexity would cost in power, security, and development effort.**

I. Introduction

II. Related Work

Radio frequency identification (RFID) has been widely studied as part of the broader class of automatic identification technologies. Kaur et al. provides a comprehensive overview of its principles, capabilities, and limitations. RFID systems enable the wireless transmission of identification using radio waves, allowing individuals or objects to be identified without requiring line-of-sight interaction, in contrast to traditional barcode systems [1]. This capability significantly improves automation and reduces the need for human action within lock systems, making RFID useful for embedded and access control applications that require high speed and reliability.

A major focus of prior work is the distinction between passive and active RFID systems, as well as the physical mechanisms that enable their operation. Passive RFID operates by harvesting energy from the electromagnetic field generated by a reader. In near-field systems, this process is governed by magnetic induction, where an alternating magnetic field induces a voltage across a tag's coil, which provides enough energy for the tag to transmit its stored ID [1]. This eliminates the need for an onboard power source. These characteristics make passive RFID well-suited for compact, power-constrained embedded systems, such as our intended implementation within a firearm's firing control group.

Despite these advantages, Kaur et al. also identifies several important limitations of RFID systems that must be addressed. One key constraint is the relationship between operating frequency and read range. In near-field systems, the effective communication range is limited and decreases rapidly due to the inverse cubic decay of the magnetic field [1]. While this restricts long-range communication, it can be advantageous for security sensitive applications by ensuring that the authenticated user is within close proximity. Other challenges include signal collision when multiple tags are present, environmental interference caused by conductive materials or electromagnetic noise, and the lack of universal standardization across RFID systems [1]. Security and privacy concerns are also significant, as unauthorized cloning of tags can compromise system integrity if appropriate safeguards are not implemented.

Taken together, the existing literature establishes RFID as a mature and well-understood technology for identification and access control. It has demonstrated advantages in automation, robustness, and low power operation. At the same time, it highlights key technical and security challenges that must be carefully addressed in our implementation. This paper provides a strong foundation for the proposed system, informing both the design choices and the tradeoffs involved in applying RFID to firearm safety.

III. System Architecture

A. Overview

The Smart Trigger is a self-contained, battery-powered embedded system that adds an electronic authorization step to the mechanical operation of an AR-15. Functionally, it is a closed loop with three stages: Sense, Decide, Act.

B. Subsystems

C. Design Requirements

The design has several requirements and constraints to meet. The priority and explanation is as follows:

1) Widely and easily applicable
   - The design is meant to work with any standard AR-15 and should be able to added in simply and easily. If it was a one off product the ultimate effect would be far less.
2) Battery operated
   - This is a simple requirement but it means that the AR-15 can be away from an outlet and be used at a gun range, hunting in remote areas, or in the field by police officers.
3) Small
   - Having a large cumbersome device added to a rifle such as the AR-15 creates a big discouragement from use. Keeping the device small and out of the way allows a user to basically hide that it even exits.
4) Reliable
   - A tool that can only be used 50% of the time is not a tool worth having. In many applications an AR needs to be used at the drop of a hat and needs to be relied on otherwise mortal consequences may ensue.
5) Quick
   - The best example is an police officer needing their rifle while in a gun fight. If the AR is not ready when the officer is it may as well not be there at all.

D. Key Architectural Decisions

IV. Hardware Components and Integration

A. Mechanical Interaction

The mechanical design has three parts:

1) a **custom grip** that replaces the factory AR-15 grip and houses the electronics, battery, and RFID antenna
2) an **actuator mount** that locates the micro stepper in the fire control pocket of the lower receiver
3) a **blocking member** driven by the motor that physically prevents the fire control group from releasing the hammer

To understand the innovation behind the blocking member takes some understanding of the original function of the AR-15 fire control group(trigger, hammer, sear, and safety). To fire a round the trigger must be pulled, the angle the trigger is brought to in its movement releases the hold it had on the spring loaded hammer. The freed hammer slams against onto a hook called the disconnector which holds the hammer until the trigger is back to its ready position it was in before firing the last round. The safety on the AR-15 is a physical barrier to the trigger. When is the safe position the trigger is physically blocked from movement thus preventing the hammer from moving and the overall firing of the gun. The design of our safety takes advantage of the mechanical blocker of the AR-15's native safety selector. To create the most widely applicable design we focused on locking the native safety in place until an authorized user is recognized.

B. Component Selection

TABLE I
Components in the Smart Trigger system

Part Number	Short Name	Short Description
Pi Pico W	Pico	Microcontroller and central brain
SAA1042	Motor Driver	Stepper motor driver
RDM6300	RFID Reader	RFID reader and translator
ANT1	Antenna	Custom wound antenna tuned to 125 kHz
M1	Motor	Stepper motor to rotate shaft
R1	R	47 kΩ resistor
C1	C	100 nF capacitor

C. Main Board (PCB)

1) Rationale for a Custom PCB: The project imposed a strict size constraint on the electronics enclosure: the usable inner cavity of the pistol grip measures 88.8 mm from top to bottom, 21.9 mm wide, and 45.0 mm across. A custom printed circuit board (PCB) would integrate all components listed in Table I into a compact form. A custom design also offered greater control over the system, since unnecessary components could be removed and only the required circuitry selected from each of the commercial development boards. Finally, a single integrated board would simplify installation for the end user compared with a multi-board assembly.

2) Design Process and Change in Approach: With these constraints in mind, we began by reverse engineering the RDM6300 RFID module, which serves as the interface between the Raspberry Pi Pico microcontroller and the RFID tag. We recreated the circuit in KiCad, a PCB design tool, and started to validate it through simulation in LTSpice. The resulting design is shown in Figure 1.

This design would have allowed us to remove the RP2040 from its development board and mount it directly on the custom PCB, eliminating the many GPIO pins we did not intend to use.

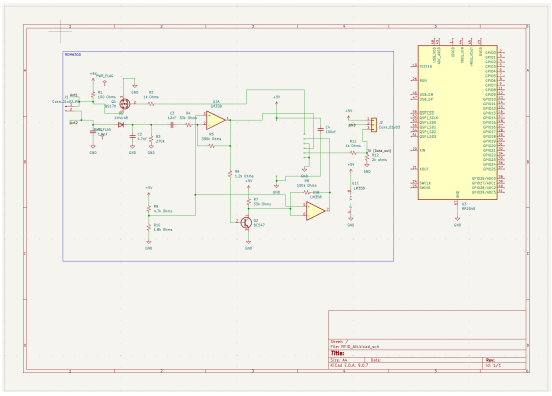


Fig. 1. First Iteration of the Custom PCB Design. The design integrates the RDM6300 RFID module, the SAA1042 stepper motor driver, and the Raspberry Pi Pico microcontroller into a single board.

Before fabricating and testing the board, however, the team questioned whether the fine-pitch leads of the RP2040 could be reliably hand soldered. Commercial manufacturers typically place such components using processes such as stencil-based solder paste application with reflow, or thermocompression bonding, neither of which was available to us. Given the project schedule and the risk of failing to mount the chip, we elected to pursue a simpler design.

3) Tradeoffs of the Revised Design: Although each of the existing boards fit within the grip, they would not be held securely in place. We therefore changed the role of the PCB to that of a perfboard-style central hub that interconnects the individual boards. This decision involved several tradeoffs. The perfboard approach retains unused features, including GPIO pins, clock circuitry, and UART pins. The boards must also be arranged to fit within the grip, making the assembly less compact than a custom PCB that integrates their functions. However, this approach required no additional parts, since all components were already in hand, and its lower complexity improved the likelihood of delivering a working product by the deadline.

The current design is shown in Figure . The board is configured to sandwich the Raspberry Pi Pico microcontroller on one side with the motor driver on the opposite side. The RDM6300 sits directly below the motor driver and is mounted through solder points that accept the short portion of its header pins, forming a compact stack of boards.

D. RFID Board (RDM6300)

1) How RFID Technology Works: RFID (Radio Frequency Identification) requires two components to communicate: a reader and a tag. The reader consists of an antenna and a transceiver, while the tag consists of an antenna and a transponder [1]. Tags are classified as either active or passive. An active tag contains a battery and can transmit its signal over long distances, whereas a passive tag has no battery and can transmit only when it is within range of a reader. The reader's antenna emits a radio frequency signal that powers the passive tag, and the strength of this signal depends on the number of coil turns in the antenna. Once powered, the tag transmits its unique ID back to the reader. The reader then decodes this signal and forwards it to a microcontroller for processing.

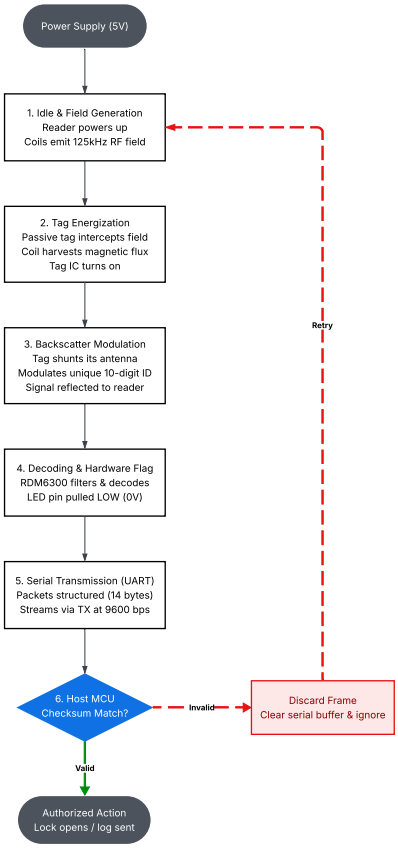


Fig. 2. A flow diagram of the RDM6300 RFID module. The module reads the RFID tag and translates its signal into a UART serial output for the Raspberry Pi Pico.

2) How we implemented RFID in our system: This project uses a passive tag because the system does not require long-range communication, and passive tags are lower in cost and power consumption. The reader antenna is custom wound and tuned to 125 kHz. The accepted tag IDs are currently hardcoded in the firmware. When a tag enters the reader's range, the RDM6300 module reads the tag and converts its signal into a UART serial output, which is sent to the Raspberry Pi

Pico. The Pico compares the received tag ID against the list of accepted IDs. If the ID matches, the Pico signals the motor driver to unlock the safety. If the ID is not accepted, no action is taken and the safety remains locked. This process is illustrated in Figure 2.

E. Micro Stepper Motor & Driver Board (SAA1042)

Stepper motors are a type of brushless DC motor that rotate when current is driven through their windings. Unlike a standard DC motor, which rotates continuously when a current is applied, a stepper motor rotates in discrete angular increments, or steps. Stepper motors are generally divided into two types. Permanent magnet stepper motors produce rotation through the interaction between a permanent magnet and energized electromagnets. Variable-reluctance stepper motors, in contrast, contain no permanent magnet and rely solely on electromagnetic forces to produce rotation.

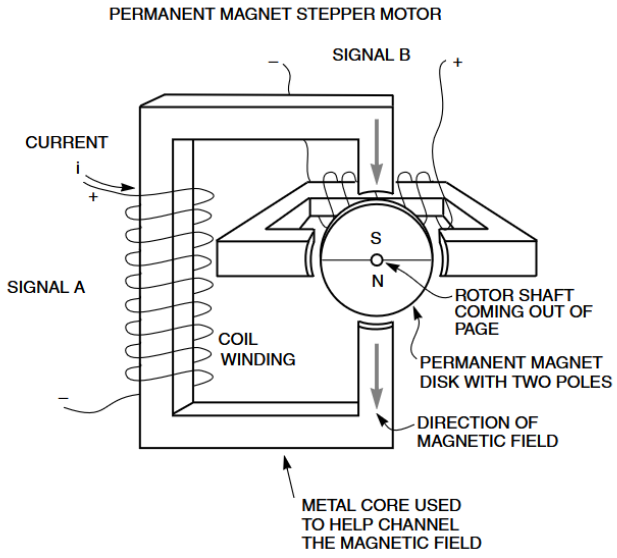


Fig. 3. A flow diagram of the stepper motor and driver board. Taken from [2]

Each motor consists of a stator, a rotor with a shaft, and coil windings, as shown in Figure 3. When current is passed through the coil windings, it generates a magnetic field that interacts with the rotor and causes it to rotate. The direction of rotation is determined by the direction of the current through the windings. The motor used in this design is a [Insert motor type here] stepper motor.

V. CAD Modeling and Design

A. Scope of the Mechanical Design

The mechanical side of the project is focused on the pistol grip, since that is where all of the electronics for the Smart Trigger will live. Fitting everything inside a grip gives us a very tight space to work with. The usable inner cavity of the grip measures 88.8 mm from top to bottom, 21.9 mm wide, and 45.0 mm long. To put that in perspective, the Raspberry Pi Pico W alone is 51 mm long and 21 mm wide, which leaves less than a millimeter of room on either side of it in the width direction and means it can only be oriented running from the top of the grip to the bottom. Every board, the motor, the antenna, and the wiring between them all have to share this space.

With that in mind, the mechanical design is built around three main goals: size, secure mounting, and usability. All of the electronics, including the Pico W, the RDM6300 RFID reader, the SAA1042 motor driver, the micro stepper motor, and the antenna, must fit within the 88.8 mm $\times$ 21.9 mm $\times$ 45.0 mm inner cavity of the grip. Fitting the boards inside the grip is not enough on its own, as they also need to be held firmly in place so they are not sliding around. A firearm sees a lot of shock and vibration from recoil and normal handling, so the mounting has to hold up even after extensive use of the gun. A loose board can break solder joints, pull wires, or shift the antenna away from where it needs to be to read the tag. At the end of the day, the grip also still has to be a usable and comfortable grip that mounts to a standard lower receiver the same way a factory grip does.

B. Custom Grip

Instead of trying to fit our electronics into an off-the-shelf grip, we chose to design and 3D print our own. Designing the grip ourselves gives us full control over the feel, grip, and appearance of the final product. We can shape the inside around our boards rather than shaping our boards around someone else's grip. We can also adjust the outside profile so it stays comfortable in the hand even with a hollowed out core. This also lets us place the RFID antenna exactly where we want it, close to where the user's hand wraps around the grip so the ring tag stays within read range.

The other big benefit of a custom grip is how easy it is to iterate on. Because we are printing the grips ourselves, a new version can go from a change in the model to a part in our hands within the same hour. This means we can test fit, find a problem, fix it, and print again without waiting on any outside vendor. For a project with as many tight tolerances as this one, being able to prototype quickly is a huge advantage.

C. Actuator Mount and Blocking Member

As described in the Hardware Components and Integration section, our blocking method works by locking the AR-15's native safety selector in place until an authorized user is recognized. The stepper motor will be housed inside the grip and will rotate a shaft that holds the safety selector in the safe position, then releases it once a valid tag is read. Because the safety selector already sits right above the grip on the lower receiver, the grip is a natural place to mount the motor close to it.

For now, the exact design of the actuator mount and blocking member is on hold until our PCB arrives. Once we have the board in hand, we will connect everything together and measure how the full assembly fits inside the grip. Those measurements will tell us how much room is left for the motor and where it can be placed, so that the mount and blocker can be designed around the real assembly instead of estimates.

D. Materials and Print Parameters

All of our grips are printed in PLA. PLA is cheap and easy to print, but it is still resilient enough for the handling and fit testing we are doing at this stage of the project. Since we expect to go through many versions of the grip before landing on a final design, keeping the cost of each print low matters.

E. Design Iteration

Our grip design started from a standard AR-15 grip model, shown in Fig. 4. This model came in as a triangle mesh, which is good for printing but difficult to edit. Making changes like hollowing out the inside or reshaping a surface is tedious when the model is made up of thousands of small triangles rather than clean faces.

For our newest iteration, shown in Fig. 5, we first converted the mesh into a solid body with clean, editable faces. With an editable model, we then extruded the middle section of the grip outward. This gives us extended room inside the grip to mount the board, and it also creates a flat surface on the exterior where the RFID antenna is mounted. Putting the antenna on the outside of the grip keeps it as close as possible to the user's hand and the ring tag, while keeping the inside free for the electronics.

The ideal final iteration will have housing for the boards and the motor built directly into the inside of the grip. Rather than having the electronics sit loosely in a hollow cavity, the inside of the grip will have slots and supports shaped to the board stack and the motor so that each part is held in a fixed position. This will keep everything secure through recoil and handling, keep the antenna in a consistent spot for reliable tag reads, and make assembly simple, since each part will only fit one way. Getting to that point depends on the measurements we will take once the PCB arrives, and with prints only taking about an hour, we expect to work through several versions quickly once we have them.

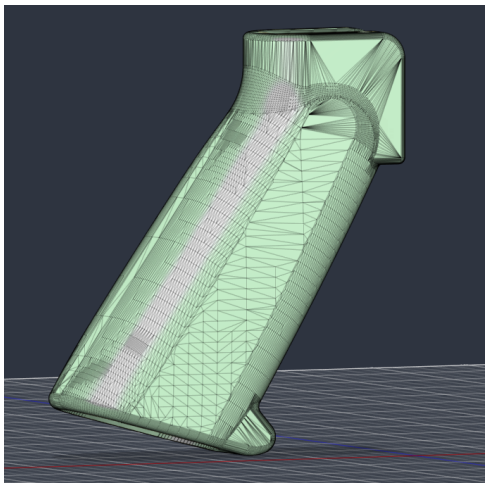


Fig. 4. The original AR-15 grip model we started from. The model is a triangle mesh, which made it difficult to modify.

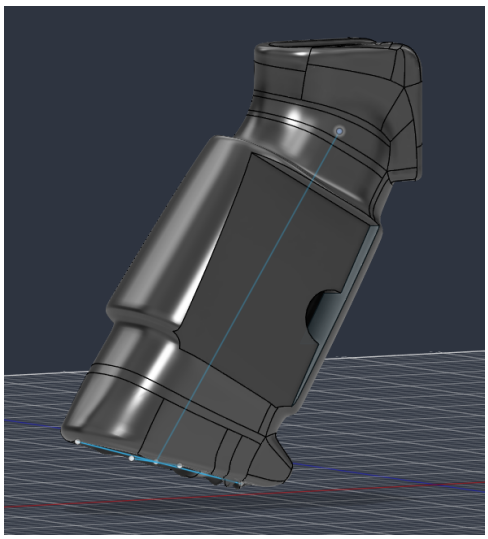


Fig. 5. Our newest grip iteration. The model was converted to a solid body and the middle section was extruded outward to make room for the board inside and the RFID antenna on the exterior.

VI. Software Processes

A. Development Environment

The development environment for the firmware is based on the Raspberry Pi Pico SDK (version 2.0 or greater), which provides a comprehensive set of tools and libraries for programming the RP2040 microcontroller

on the Raspberry Pi Pico W. The firmware is written in C11 and utilizes the FreeRTOS SMP kernel with Heap 4 memory management to manage tasks and ensure responsive, deterministic performance. Build configuration and cross-compilation are handled via CMake, targeting both USB serial and an independent hardware UART debug interface.

B. Firmware Architecture

The firmware architecture is designed to be modular and maintainable, with each component having a specific role and responsibility divided across isolated compilation units (‘main.c’, ‘rfid.c’, and ‘stepper.c’). The system is divided into several layers, including the hardware peripheral access layer (Pico SDK hardware drivers), an RTOS task synchronization layer, and the application logic layer. This design allows for easier debugging and maintenance while ensuring that the system remains robust under typical usage scenarios by preventing driver operations from blocking critical timing loops.

C. Task Management and Inter-Task Synchronization

Rather than relying on a polling super-loop, the firmware implements four concurrent FreeRTOS tasks to manage system states and hardware operations: ‘rfid_task’, ‘stepper_task’, ‘led_task’, and ‘console_task’. Inter-task communication and hardware decoupling are handled through kernel synchronization primitives:

- UART Interrupt and Stream Buffer: A dedicated UART interrupt service routine (‘rfid_uart_irq’) drains incoming reader bytes directly from the 32-byte hardware FIFO into an RTOS stream buffer (‘rx_stream’), eliminating byte drops caused by serial print latency in application tasks.
- Task Notifications: When ‘rfid_task’ validates an authorized tag, it signals ‘stepper_task’ using lightweight FreeRTOS direct task notifications (‘xTaskNotifyGive’ and ‘ulTaskNotifyTake’) via ‘stepper_start()’, and halts rotation with ‘stepper_stop()’ when the tag is removed.
- Decoupled Actuator and LED Scheduling: ‘stepper_task’ signals ‘led_task’ on each completed step pulse via task notifications, decoupling the timing of wireless chip I/O from physical motor control.

D. RFID Processing

The RFID processing component is responsible for reading and interpreting data from an external RDM6300 125kHz RFID module operating at 9600 baud (8N1) over UART0 on GP1. It handles communication with the reader, processes incoming serial bursts, and determines appropriate actions based on tag information:

- Frame Demarcation: The parser buffers data bounded by an STX (0x02) start byte and an ETX (0x03) end byte, capturing a 10-character ASCII hex payload and a 2-character checksum.
- Checksum Verification: The module verifies the RDM6300 XOR checksum across the five decoded data bytes before any access evaluation occurs.
- Authentication and Presence Timeout: The decoded tag is matched against a pre-compiled identifier macro (‘RFID_VALID_TAG’, defaulting to EM410x tag ‘BEAD101000’). If valid, the system maintains trigger activation while tag frames continue to arrive, halting rotation if no valid frame is received within a 500 ms timeout window (‘RFID_GONE_MS’) or if an unauthorized tag is detected.

E. Actuator Control

The actuator control component manages the physical operation of an Abovehill 8 mm micro stepper motor driven by an SAA1042 stepper controller IC in half-step mode (20 full steps/rev). It receives commands from ‘rfid_task’ and translates them into specific actions:

- Pulse Generation: ‘stepper_task’ outputs a 10 ms active-high step pulse (‘STEP_PULSE_WIDTH_MS’) on GP22.
- Deterministic Periodicity: Pulse cadence is locked to a 100 ms interval (‘STEP_PERIOD_MS’) using ‘vTaskDelayUntil’, providing consistent mechanical stepping without timing drift.
- Directional Control: Direction is set to clockwise (‘STEPPER_TRIGGER_CW’) by asserting GP21 prior to stepping.

F. Hardware Interface and Pad Configuration

The firmware directly addresses the physical electrical constraints of the peripheral components:

- 5V to 3.3V Logic Interfacing: Because the RDM6300 module transmits at 5V and RP2040 GPIO pads tolerate a maximum of 3.3V, serial data is received through a 2kΩ series resistor on GP1, with dynamic software pull-downs configured for connection diagnostics.
- Slew Rate and Pad Drive Strength: To ensure clean switching thresholds across the SAA1042 input logic, GP21 (direction) and GP22 (clock) are programmed for fast slew rates and an increased 12 mA drive strength.

- CYW43 Bus Isolation: On the Raspberry Pi Pico W, the onboard status LED is driven via the Infineon CYW43439 wireless chip over a shared internal bus. Because communicating with the CYW43 involves SPI transactions that introduce timing latency, LED state changes are isolated to a dedicated low-priority background task (‘led_task’), preventing motor step jitter.

G. Software Testing and Hardware Diagnostics

Testing and diagnostic instrumentation are integrated directly into the firmware architecture:

- Integrated Self-Test Modes: Using the ‘RFID_SELFTEST’ build flag, the firmware can run full automated regression tests without physical tags. Mode 1 activates the internal hardware UART loopback inside the RP2040 (‘UART_UARTCR_LBE_BITS’), while Mode 2 configures GP0 as an external transmitter jumpered to GP1 to test pad and trace continuity. A synthetic test task (‘rfid_test_task’) transmits alternating valid and invalid framed packets every 2000 ms to verify checksum logic and driver triggering.
- Line State and Bus Diagnostics: A periodic diagnostics routine polls the physical line state of GP1 and reports hardware-level UART errors, tracking framing errors, line breaks, and FIFO overruns on the debug console.
- Telemetry Console: Diagnostic logging is broadcast over a dedicated secondary UART (UART1 on GP4 at 115200 baud) and concurrent USB CDC serial (‘stdio_usb’), isolating debug telemetry from the RFID module interface.

H. Future Work: Networking and Telemetry Integration

The Smart Trigger system architecture is prepared for expansion into network telemetry and centralized access auditing using the onboard CYW43439 wireless chip of the Raspberry Pi Pico W. While the present implementation links ‘pico_cyw43_arch_sys_freertos’ strictly for onboard LED control with the lwIP stack disabled (‘CYW43_LWIP=0’), the underlying FreeRTOS environment is already structured to support networking. Transitioning to an active network node requires:

1) Enabling the lwIP TCP/IP stack (‘CYW43_LWIP=1’) within the CMake build target to enable socket-level networking.
2) Implementing a secure FreeRTOS client task (using MQTT over TLS or HTTPS) to publish access logs, timestamped valid reads, and tamper/break alerts to an offloaded database server.
3) Deploying a web service dashboard to authenticate administrators and permit remote over-the-air updates to the authorized tag whitelist (‘RFID_VALID_TAG’).

VII. Conclusion

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[2] M. Grant, "Quick start for beginners to drive a stepper motor," *Freescale Semiconductors),[online]*, vol. 6, 2005.